Quark Matter Phase Boundary Evolution in Neutron Star Mergers

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I. INTRODUCTION

At sufficiently high densities, nuclear matter is theorized to undergo a phase transition to deconfined quark matter. While the exact nature of the phase transition is unknown, a natural possibility includes a sharp interface between nuclear matter and quark matter due to a first-order phase transition. In this scenario, the phase transition occurs at a (unknown) critical pressure with a discontinuity in the energy density , determining the location and strength of the phase transition. Such a transition might occur in several astrophysical scenarios, ranging from core-collapse supernovae to isolated neutron stars and binary neutron star mergers. By observing signatures of the first order phase transition in these systems in the gravitational or electromagnetic waves emitted, we are aiming to explore the phase structure of dense matter.

During inspiral, a phase boundary between nuclear and quark matter may support interface modes whose excitation produces signatures that are potentially within reach of next-generation detectors like Cosmic Explorer or the Einstein Telescope [1, 2].

In the post-merger phase, there are also possible signatures of a nuclear-quark phase boundary. Several recent works have addressed this question using model equations of state that include a nuclear/quark transition [3–12], looking for features of the gravitational wave emission arising from the redistribution of mass as regions of the denser phase grow or shrink in response to pressure gradients in the violent collision between the stars.

The same phase transition may also power core-collapse supernovae: if the collapsing core of a massive star reaches the density where nuclear matter converts to quark matter, the resulting quark core can trigger a delayed second shock and explode stars that neutrino heating alone fails to explode [13–15].

In isolated neutron stars, spin-down-induced compression may lead to the formation of a quark-matter core or to the growth of an existing one [16, 17]. Depending on the conversion dynamics, this growth may proceed gradually or through discrete episodes involving metastable hadronic matter, potentially producing observable timing irregularities such as pulsar glitch-like events [18].

In all of these scenarios, the key question is how fast a region of metastable nuclear matter could be converted to quark matter via movement of an existing phase boundary arising from a gradual (compared to microscopic timescales) increase in pressure. The velocity with which an existing quark matter region can grow by converting nuclear matter to quark matter. This phase boundary velocity has a profound impact on the physical outcome and observational signature. However, in all of the cited studies, it has been treated in one of two extreme limits: the zero or infinite velocity limit. [AH] I am not sure if we should explain for all of these scenarios why the phase boundary velocity matters or how. I feel like it would make it too long and derail the message. Maybe we should do this in the results section or final discussion and give an estimate using our results. The old explanations are commented out in the right new place if we want to add them

The propagation of a nuclear/quark phase boundary has been analyzed before, mostly for the more extreme case of the conversion of hadronic stars into strange stars in the context of the strange matter hypothesis [[MGA] For now I'm not separating ones that allow for a pressure-induced transition] [19–29] This paper complements earlier work by studying a slowly varying pressure driven transition to quark matter (rather than the global instability assumed by the strange matter hypothesis), studying the temperature and density ranges typical of neutron star mergers and supernovas, giving a survey of how the boundary speed varies as a function of the density and temperature of the nuclear phase, and providing an approximate analytic expression for the boundary speed. To achieve this, we adopt a simplified model, focusing on the dominant flavor equilibration channel in quark matter. We do not consider the reverse process, conversion of quark matter to nuclear matter, although this may also occur in situations where there are pressure oscillations. We aim to set an approximate upper bound on the propagation speed of the boundary, so our approximations and simplifications are chosen to favor fast propagation.

Our treatment assumes a deflagration-type conversion process, in which the pressure has risen high enough so that nuclear matter becomes metastable relative to 3-

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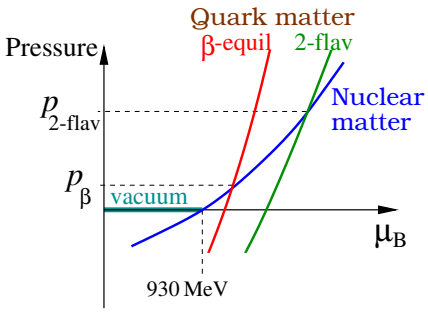


FIG. 1: Pressure comparison between phases at some arbitrary temperature. Above pressure P_β, 3-flavor chemically equilibrated quark matter is favored over nuclear matter. At $P > P_{2\text{-flav}}$, quark matter with the same flavor content as nuclear matter becomes favored, so weak interactions are no longer required for the conversion of nuclear to quark matter.

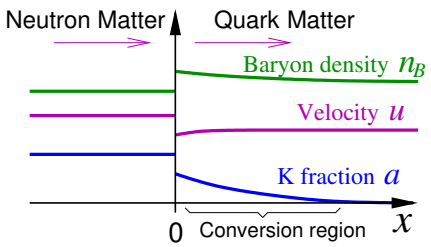


FIG. 2: Schematic illustration of the expected boundary profile. Note the discontinuities in some thermodynamic properties at the phase boundary located at $x = 0$, followed by relaxation to beta equilibrium as $x \to \infty$. In the adiabatic case, temperature will also show a discontinuity at $x = 0$ followed by relaxation to an asymptotic value as $x \to \infty$.

flavor quark matter, so weak interactions are required for the phase boundary to move. There is a more extreme possibility, only realized if the pressure becomes so high (e.g. in an externally induced shock front) that quark matter with the same flavor content as nuclear matter is energetically preferred. This opens up the possibility of a detonation-type conversion [26, 30] that does not require weak interactions and hence can proceed much faster. AH I think there is a scenario more likely in between: 2 flavor quark matter being favored, but still different flavor content, then upness needs to be created and diffused via Urca. I don't think we should go into that detail here, but mention the possibility?

AH not sure how the next paragraph fits in but the citations are important A more violent version of this process, the quark-nova, is the hypothesized runaway conversion of an entire neutron star into a quark star, invoked as an energy source for superluminous supernovae and other high-energy transients [23, 29]. The microphysical boundary speed derived here is the starting point for such global conversion scenarios; extending our steady, planar treatment to the turbulent, whole-star regime of a quark-nova is left for future work.

The paper is structured as follows. In Sec. II, we present our assumptions and the hydrodynamic equations in the steady-state approximation. Sections III and IV describe the isothermal and adiabatic calculations of the boundary speed, respectively. In Sec. V we present our results, and in Sec. VI we summarize our conclusions and describe directions for future work. Our calculations use natural units, $\hbar = c = k_B = 1$, and metric signature $(+ - - -)$.

II. FORMALISM AND ASSUMPTIONS

A. Main assumptions

We assume the phase boundary is well approximated as a flat plane which can therefore be modeled as a one-dimensional system. This is a reasonable approximation for neutron star mergers, where the curvature of the boundary is on the kilometer scale, while its thickness is of order micrometers (see Eq. (37)). We defer any discussion of instabilities that would lead to a more complicated front geometry. AH As far as I know, people assume that instabilities work on the cm scale, which would still give us many orders of magnitude compared to micrometer. So the planar assumption should be fairly solid We assume that the surface tension of the phase boundary is large enough that mixed phases are not favored. We model the nuclear phase as pure neutron matter described by the QMC-RMF3 mean field theory [31–33]. We model quark matter as a Fermi gas of massless noninteracting up, down, and strange quarks [34, 35].

We will assume that the phase boundary is at a pressure where, in the range of temperatures relevant to neutron stars, 3-flavor beta-equilibrated quark matter is favored over nuclear matter (approximated as neutron matter in our case), but 2-flavor quark matter with the same flavor content as nuclear matter is not favored. I.e., the ambient pressure P is between P_β and $P_{2\text{-flav}}$ in Fig. 1. Thus weak interactions are required to complete the conversion of the metastable nuclear matter to quark matter. (In the adiabatic case the comparison is more complicated because the two phases are at different temperatures, see Sec. IV). AH So only because we model nuclear matter as neutron matter, the flavor content is exactly ud in ration 1:2, just like two-flavor charge neutral QM. But chemically equilibrated nuclear matter and two-flavor QM will have a different flavor content, and this will probably lead to an even slower speed because we now need Urca. I think this should be discussed? We will follow the standard approach to nuclear to quark conversion [19, 27, 28, 36–

38] which exploits the clear separation between strong-interaction-driven processes such as local thermal equilibration and deconfinement, which can be treated as instantaneous on the time scales relevant to neutron star mergers, and flavor equilibration, which occurs via weak interactions and is therefore much slower and must be treated explicitly.

We focus on the fastest flavor equilibration channel, the nonleptonic process

$$d\ u \leftrightarrow u\ s\ . \tag{1}$$

This equilibrates the quantum number that we will call "K", which measures the imbalance between down and strange quarks. We ignore electromagnetism, electrons, and neutrinos. We therefore neglect Urca processes, which are generally slower than the non-leptonic process [39–41], so their inclusion will make the boundary speed even lower.

As noted above, we assume local thermal equilibration is fast enough so that each fluid element has a well-defined temperature T. We will first write down the hydrodynamic equations that describe the movement of the fluid, including relativistic corrections, ~~radiative loss~~ [MGA Not in the hydro eqns any more] and diffusion of flavor (the K quantum number) and heat. In the body of the paper, however, we will self-consistently neglect relativistic hydrodynamic corrections since we will find that the boundary moves at speeds far less than the speed of light. Moreover, we will not explicitly treat the radiation or diffusion of heat via thermal conduction. Instead, we will calculate the boundary propagation speed for two extreme cases: isothermal (the whole system is thermally equilibrated to the same temperature) and adiabatic (no thermal conduction).

B. Formalism

To calculate the speed at which the nuclear/quark phase boundary propagates through neutron matter, we will look for a steady state solution to the equations that describe the propagation of a deflagration front.

We work in the rest frame of the phase boundary, which is located at $x = 0$, as illustrated schematically in Fig. 2. At $x < 0$ there is pure neutron matter which we take to be homogeneous, neglecting any disturbance propagating ahead of the phase boundary. The neutron matter flows towards the boundary with a velocity $u(x = 0^-)$, which we aim to calculate. Matter at $x = 0^-$ immediately deconfines into denser quark matter at $x = 0^+$ with enough strangeness to be created from neutron matter under the prevailing conditions [MGA These weasel words are meant to cover the isothermal vs adiabatic vs everything in between]. AH I don't understand the enough strangeness part? [MGA If the quark matter doesn't have enough strangeness then it won't be energetically favored (extreme case: udd matter).]

At $x > 0$ there is a conversion region where strange quarks are created via the weak interaction, leading to a flavor gradient that drives diffusion of strange quarks upstream towards the boundary. This is how the freshly created quark matter at $x = 0^+$ can immediately have a larger strangeness fraction than was present in the incoming neutron matter. As $x \to \infty$ the quark matter approaches a fully beta-equilibrated state.

The parameters that describe the state of the matter at any given position x are:

$$\begin{array}{ll} n_B & \text{baryon number density} \\ n_K \equiv \dfrac{n_d - n_s}{2} & \text{"K" density} \\ u & \text{velocity in the } x \text{ direction} \\ T & \text{temperature} \end{array} \tag{2}$$

From these one can, via the nuclear and quark equations of state (see Appendix A), calculate other properties such as pressure P, baryon chemical potential μ_B, energy density, and the K chemical potential

$$\mu_K = \mu_d - \mu_s\ , \tag{3}$$

which relaxes towards zero via the weak interaction (1).

We will find it useful to define the local K fraction

$$a(x) \equiv \frac{n_K(x)}{n_B(x)}\ . \tag{4}$$

C. Hydrodynamic equations

Although our calculations will focus on the isothermal and adiabatic regimes, we first present the more general hydrodynamic equations including the effects of flavor and heat diffusion and radiative energy loss but neglecting viscous stresses.

The stress-energy tensor without viscosity terms takes the form [42, 43]

$$T^{\mu\nu} = (P + \epsilon)U^\mu U^\nu - P g^{\mu\nu} + U^\mu q^\nu_{\rm th} + q^\mu_{\rm th} U^\nu\ , \tag{5}$$

and the fluid flow is defined by the baryon current

$$J^\mu_B = n_B U^\mu\ , \tag{6}$$

where $U^\mu = (\gamma, u_x, u_y, u_z)$ is the baryon fluid four-velocity, with Lorentz factor $\gamma = 1/\sqrt{1 - v^2}$, spatial component of the four-velocity $u = \gamma v$, and lab-frame three-velocity v, and $q^\nu_{\rm th}$ is the heat current which is purely spatial in the local fluid frame, i.e., $U_\mu q^\mu_{\rm th} = 0$.

We now apply the relevant conservation laws. Baryon number is conserved, while radiation can carry energy and momentum out of the fluid: [MGA If we are going to neglect radiation in eq (16), is there any point in keeping it here?][LB I am fine without Q^ν. Alex wanted this in the past so lets ask him.]

$$\begin{aligned} \partial_\mu J^\mu_B &= 0\ , \\ \partial_\mu T^{\mu\nu} &= -Q^\nu\ . \end{aligned} \tag{7}$$

Here Q^ν is the radiation energy-momentum loss per unit spacetime volume. There is also a K quantum number current, whose conservation is violated by weak interactions,

$$\partial_\mu J_K^\mu = -\Gamma_K \ , \tag{8}$$

where Γ_K is the rate of down-to-strange quark conversion (see Appendix B) and

$$J_K^\mu = n_K U^\mu + D_K^\mu \ , \tag{9}$$

where D_K^μ is the K diffusion current which obeys $U_\mu D_K^\mu = 0$. The relativistic diffusion current D_K^μ can be constructed in general by projecting the thermodynamic driving force (here $\partial_\nu n_K$) orthogonally to the four-velocity (see Sec. 14.1 of Ref. [44]). We use the relativistic Fick law form

$$D_K^\mu = D_K \Delta^{\mu\nu} \partial_\nu n_K \ , \tag{10}$$

where $\Delta^{\mu\nu} = (g^{\mu\nu} - U^\mu U^\nu)$ projects onto directions orthogonal to the fluid four-velocity and D_K is the K-diffusion coefficient.

We now formulate these conservation equations in the planar phase boundary rest frame. Derivatives with respect to t, y, and z vanish. Then baryon number conservation gives

$$\frac{d}{dx}(n_B u) = 0 \, , \tag{11}$$

and energy-momentum conservation reduces to

$$\frac{d}{dx}\left[(P+\epsilon)\gamma u + \frac{\gamma^2+u^2}{\gamma} q_{\rm th}^x\right] = -Q^0 \, , \tag{12}$$

$$\frac{d}{dx}\left[(P+\epsilon)u^2 + P + 2u q_{\rm th}^x\right] = -Q^x \, . \tag{13}$$

Note that we have used $q_\mu U^\mu = 0$ and $\gamma^2 - u^2 = 1$.

For the heat current $q_{\rm th}^\mu$ we use the Eckart heat law [42, 45]

$$q_{\rm th}^x = -\kappa_{\rm th} \gamma^2 \frac{dT}{dx} \, , \tag{14}$$

where $\kappa_{\rm th}$ is the thermal conductivity [46].

In our 1-dimensional geometry we write the spatial component of the K current as

$$J_K = u n_K - \gamma^2 D_K \frac{d}{dx} n_K \ . \tag{15}$$

See Appendix C for our approach to estimating the diffusion coefficient D_K.

Neglecting radiative effects $Q^\mu = 0$, we arrive at the following set of hydrodynamic equations

$$\frac{d}{dx}(u n_B) = 0 \, , \tag{16a}$$

$$\frac{d}{dx}\left[(P+\epsilon)\gamma u + \frac{\gamma^2+u^2}{\gamma} q_{\rm th}^x\right] = 0 \, , \tag{16b}$$

$$\frac{d}{dx}\left[(P+\epsilon)u^2 + P + 2u q_{\rm th}^x\right] = 0 \, , \tag{16c}$$

$$\frac{d}{dx} J_K = -\Gamma_K \, . \tag{16d}$$

In the following sections we will solve Eqns. 16 to obtain the propagation speed of the phase boundary u in the isothermal and adiabatic limits.

The isothermal limit (Sec. III) is obtained by assuming infinite thermal conductivity, so the system will have a uniform temperature.

The adiabatic limit Sec. IV) is obtained by setting the thermal conductivity to zero, so entropy is conserved, apart from contributions from the weak interactions that relax the system towards beta-equilibrium[LB check this]. The steady-state boundary profile will then have a spatially varying temperature.

Anticipating the results that will be presented in Sec. V, which find that the typical boundary propagation speeds are around a millionth of the speed of light, we will ignore relativistic effects from now on by setting $\gamma = 1$ and dropping $\mathcal{O}(u^2)$ corrections, so we no longer distinguish the three-velocity v from the x component $u = \gamma v$ of the four-velocity.

III. BOUNDARY PROPAGATION SPEED: ISOTHERMAL CASE

A. Isothermal hydrodynamic equations

The isothermal limit is obtained by taking $\kappa_{\rm th} \to \infty$ in Eqns. 16. This yields

$$\frac{d}{dx}(u n_B) = 0 \, , \tag{17a}$$

$$\frac{d}{dx} T(x) = 0 \, , \tag{17b}$$

$$\frac{d}{dx} P(x) = 0 \, , \tag{17c}$$

$$\frac{d}{dx} J_K = -\Gamma_K \ , \tag{17d}$$

where we obtained Eq. (17b) from the $\kappa_{\rm th} \to \infty$ limit of Eqs. (16b) (or (16c)) combined with (14). Eq. (17c) is obtained by writing Eqs. (16b) and (16c)) as $(P+\epsilon)u + q_{\rm th} = C_1$ and $P + 2u q_{\rm th} = C_2$ where C_1 and C_2 are independent of x, and eliminating $q_{\rm th}$ and dropping terms of order u^2 to obtain $P = C_2 - 2uC_1$. Using the fact that $q_{\rm th} \to 0$ at $x \to \infty$ (because all gradients including dT/dx vanish) we can see that C_1/C_2 is of order u, so again dropping terms of order u^2 we find that P is independent of x.

The first three equations give us simple constancy conditions, for the baryon current, temperature, and pressure. Thus to obtain the profile of the moving phase boundary we only need to solve the K flux equation Eq. (17d), a second-order differential equation that can be recast as two first-order differential equations for the

K density and current,

$$\frac{d}{dx}n_K = \frac{un_K - J_K}{D_K}\,, \tag{18a}$$

$$\frac{d}{dx}J_K = -\Gamma_K\,. \tag{18b}$$

where D_K and Γ_K in quark matter ($x > 0$) are given in Appendices C and B. We do not need to know D_K or Γ_K in nuclear matter ($x < 0$), because we assume it is uniform and equilibrated. At the boundary itself we expect Γ_K will remain finite, but the right side of Eq. (18a) will include a delta-function singularity (as the corresponding equation for dn_B/dx does, since u and n_B both have discontinuities).

There is now a well defined procedure for constructing a solution to the hydrodynamical equations. Assume we have solved Eq. (18) up to some position x, giving $n_K(x)$ and $J_K(x)$. We can combine this with the already-specified x-independent values of J_B, P, and T and the quark matter equation of state (Appendix A) at the prescribed temperature to solve for the remaining properties of the matter at x, such as $n_B(x)$, $u(x)$, $\mu_B(x)$, $\mu_K(x)$, etc. This allows us to calculate $D_K(x)$ and $\Gamma_K(x)$ (see Appendices B and C) and thereby obtain $n_K(x+\delta x)$ and $J_K(x+\delta x)$.

B. Boundary conditions

To construct the boundary profile we need 5 initial conditions specifying the state of the incoming nuclear matter: $n_B(0^-)$, $n_K(0^-)$, $dn_K/dx(0^-)$, $T(0^-)$, $u(0^-)$. However, since we assumed the incoming matter is neutron matter, for which $n_K = n_d/2$, we have

$$n_K(0^-) = n_B(0^-)\,, \tag{19}$$

and since we have assumed the neutron matter is homogeneous,

$$\frac{dn_K}{dx}(0^-) = 0\,, \tag{20}$$

so we only need to specify $n_B(0^-)$, $T(0^-)$, $u(0^-)$, which fixes the x-independent parameters J_B and P for the whole solution.

The first step in solving the equations is to relate the K density and current at $x = 0^+$ to the initial conditions at $x = 0^-$, after which we can evolve the solution to $x = +\infty$ using (18).

Integrating Eq. (18b) across the sharp interface yields the continuity condition $J_K(0^+) = J_K(0^-)$, which can be re-expressed using Eq. (18a) as a hydrodynamic jump condition,

$$u(0^+)n_K(0^+) - \left[D_K\frac{dn_K}{dx}\right]_{0^+} = u(0^-)n_K(0^-)\,. \tag{21}$$

We do not need to know the K diffusion coefficient on the nuclear side because we have assumed there is no gradient there.

We cannot integrate Eq. (18a) to obtain an analogous relation between $n_K(0^+)$ and $n_K(0^-)$, because the right side of Eq. (18a) includes an unknown delta-function divergence at $x = 0$. We therefore have to treat $n_K(0^+)$ as an additional parameter of the system. In our calculations we will choose the value of $n_K(0^+)$ that maximizes the speed of the boundary.

Since the solution must yield equilibrated quark matter as $x \to \infty$, we also impose the boundary condition

$$\mu_K(\infty) = 0\,, \tag{22}$$

which for our model of quark matter is equivalent to $n_K(\infty) = 0$. This turns (18) into an eigenvalue problem, where for given $n_B(0^-)$, $T(0^-)$, and $n_K(0^+)$ there is only one value of the incoming neutron matter velocity $u(0^-)$ that yields a valid solution.

We can now obtain the boundary speed in terms of the density and temperature of the neutron matter and the K fraction in the freshly created quark matter. We do this by solving (18) as described in Sec. III A using the boundary conditions described above.

Our results presented in Sec. V will be obtained by solving the differential equations numerically. However, as we now describe, one can also formulate an approximate analytic solution, which is helpful for interpreting the results.

C. Analytic approximation

To obtain an analytic approximate solution for the phase boundary propagation speed, we note that in the conversion region, from $x = 0^+$ to ∞, as μ_K relaxes to zero, the constancy of the pressure and temperature requires the quark matter to follow an isobar in the (μ_B, μ_K) plane as shown in Fig. 3.

From the bag model equation of state (A5), we see that $d\mu_B/d\mu_K|_P \sim \mu_K/\mu_B$ which is of order 1%, so along this isobar we expect μ_B (and consequently n_B, which is mainly determined by μ_B) will only vary by a small amount, as seen in Fig. 3. We therefore develop an analytic approach based on approximating μ_B and n_B (and therefore the velocity u, see (17a)) as constant from $x = 0^+$ to ∞, in addition to pressure and temperature, writing the x-independent baryon density in the quark matter as

$$n_B^{(+)} \equiv n_B(x) \quad \forall x > 0\,. \tag{23}$$

We define λ_n (which will always be less than 1) as the ratio of neutron matter to quark matter baryon density,

$$\lambda_n \equiv \frac{n_B(0^-)}{n_B^{(+)}}\,, \tag{24}$$

so the x-independent velocity of the quark matter is

$$u^{(+)} = \lambda_n u(0^-)\,. \tag{25}$$

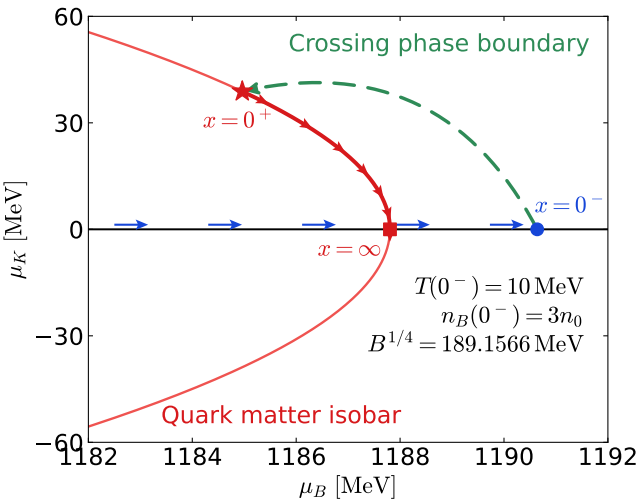


FIG. 3: Constant-pressure trajectory through the quark conversion layer in the (μ_B, μ_K) plane at fixed $T = 10\,\mathrm{MeV}$. The red curve is the quark-matter isobar for $n_B(0^-) = 3n_0$ and $B^{1/4} = 189.1566\,\mathrm{MeV}$. The red arrows run from the just-deconfined state at $x = 0^+$ to the equilibrated state at $x \to \infty$; the green arrow represents deconfinement from the nuclear-side marker at $x = 0^-$.

Since n_B is independent of x, it is convenient to work in terms of the local K fraction $a(x)$ defined in Eq. (4).

To find the speed $u(0^-)$ we use Eq. (21) along with (23), (24), (25) to obtain

$$\begin{aligned} u(0^-) &= -\frac{D_K(0^+)a'(0^+)}{\lambda_n\big(1-a(0^+)\big)}\,, \\ \text{where}\quad a'(x) &\equiv \frac{da}{dx} = \frac{1}{n_B^{(+)}}\frac{dn_K}{dx}\,. \end{aligned} \tag{26}$$

The boundary conditions determine λ_n and $a(0^+)$, but $D_K(0^+)a'(0^+)$ is determined by the requirement that $a(\infty) = 0$. To impose this exactly we would have to evolve $a(x)$ from $x = 0^+$ to infinity using the K current equation Eq. (18).

We cannot solve this equation exactly in full generality, but we can make a good estimate of the result. Multiplying Eq. (18b) by $D_K(x)a'(x)$, using Eq. (18a) to eliminate J_K, and integrating from $x = 0^+$ to infinity, we find

$$\frac{1}{2}\Big(D_K(0^+)\,a'(0^+)\Big)^2 = -u^{(+)}I_1 + I_2\,, \tag{27}$$

where I_1 and I_2 are positive dimensionless integrals over x which can be rewritten as integrals over a, since $a(x)$ is expected to be a monotonically decreasing function over $0^+ \leqslant x < \infty$ as the quark matter equilibrates in the conversion region behind the boundary.

$$\begin{aligned} I_1 &= \int_{0^+}^{\infty} D_K(x)\big(a'(x)\big)^2\,dx\,, \\ &= -\int_0^{a(0^+)} D_K a'\,da\,. \end{aligned} \tag{28}$$

$$\begin{aligned} I_2 &= -\frac{1}{n_B^{(+)}}\int_{0^+}^{\infty} D_K\Gamma_K a'\,dx\,, \\ &= \frac{1}{n_B^{(+)}}\int_0^{a(0^+)} D_K\Gamma_K\,da\,. \end{aligned} \tag{29}$$

These expressions are valid however strongly D_K and Γ_K depend on x, so we will be able to use them in the adiabatic case also.

In an exact calculation both integrands should be obtained from the solution to Eq. (18), but we will now show how one can obtain reasonable approximations to both integrals without knowing the solution.

We first estimate I_1 because the estimate will turn out to depend on our desired result $u(0^-)$, requiring regrouping of Eq. 26. We assume that, as we see in the numerical solutions, the actual profile has $D_K(x)a'(x)$ increasing monotonically from its value $-\lambda_n u(0^-)[1 - a(0^+)]$ at $x = 0^+$ (from Eq. (26)) to zero at $x = \infty$. This is expected because $a(x)$ decreases monotonically to zero, and any variation in the diffusion constant D_K, which depends mainly on the temperature, is too weak to overcome that. We can then parameterize the integral in terms of a "sharpness" factor ξ:

$$\begin{aligned} I_1 &= -\frac{1+\xi}{2}\,D_K(0^+)a'(0^+)\,a(0^+) \\ &= \frac{1+\xi}{2}\,\lambda_n u(0^-)\,[1-a(0^+)]\,a(0^+)\,. \end{aligned} \tag{30}$$

Here $-1 < \xi < 1$ characterizes the shape of the diffusion profile. For details of this sharpness factor ξ and derivation of Eq. (30), see Appendix D. Substituting this relation into (27) and (26) yields

$$u^2(0^-) = \frac{2I_2}{\lambda_n^2[1-a(0^+)][1+\xi a(0^+)]}\,, \tag{31}$$

We now evaluate I_2 in the isothermal regime. The K decay rate Γ_K is given by Eq. (B3), evaluated at the constant quark-side baryon chemical potential $\mu_B^{(+)}$. At fixed T and $\mu_B^{(+)}$, the coefficients D_K and γ_K are constant, and Eq. (29) reduces to

$$\begin{aligned} I_2 &= D_K\gamma_K\int_0^{a(0^+)}\left(a^3 + \frac{81\pi^2T^2}{(\mu_B^{(+)})^2}a\right)da \\ &= \frac{D_K\gamma_K\,a^2(0^+)}{4}\left[a^2(0^+) + \frac{162\pi^2T^2}{(\mu_B^{(+)})^2}\right]\,. \end{aligned} \tag{32}$$

Substituting Eq. (32) into Eq. (31) we obtain the analytic estimate of the phase boundary speed,

$$u_{\rm iso}^2(0^-) = \frac{D_K\gamma_K\, a^2(0^+)\left[a^2(0^+) + 162\pi^2T^2/(\mu_B^{(+)})^2\right]}{2\lambda_n^2\left(1-a(0^+)\right)\left(1+\xi a(0^+)\right)}\,, \tag{33}$$

where λ_n and ξ are defined in Eqs. (24) and (30), respectively.

The two terms shown in square brackets in the numerator arise from the cubic and linear contributions to the K decay rate in Eq. (B3). The cubic contribution dominates when $a(0^+) \gg 9\pi T/\mu_B^{(+)}$, while the thermal contribution dominates at higher temperatures or smaller $a(0^+)$.

The final step is to choose a value for $a(0^+)$, the K fraction in freshly created quark matter, which lies between 0 (corresponding to quark matter with as many strange quarks as there are down quarks), and 1 (corresponding to quark matter with no strangeness, i.e., having the same flavor content as neutron matter, $a(0^-) = 1$). We noted in Sec. III B that this is an unspecified parameter in our calculation. We will now choose it so as to give the maximum speed of the boundary $u_{\rm iso}^{\rm max}(0^-)$. Since the speed is a monotonically increasing function of $a(0^+)$ in the domain $0 < a(0^+) < 1$ for any value of the sharpness factor $-1 < \xi < 1$, this means we set $a(0^+)$ to $A_{\rm iso}$, the largest physically possible value of $a(0^+)$, which is given by the requirement that the quark matter at $x = 0^+$ must be thermodynamically favored over the neutron matter at $x = 0^-$. Since these two forms of matter are at the same pressure and temperature, neutron→ quark conversion is favored when the average chemical potential of a quark at $x = 0^+$ is lower than the average chemical potential of a quark in neutron matter at $x = 0^-$. Thus $A_{\rm iso}$ is specified as the value of $a(0^+)$ for which the average chemical potentials are equal. Quark matter with $a(0^+) > A_{\rm iso}$ would have insufficient strangeness to make it favored over neutron matter (see Fig. 1 and Ref. [25]).

Since Eq. (33) has monotonic dependence on $a(0^+)$ and $0 < a(0^+) < A_{\rm iso} < 1$, setting $a(0^+) = A_{\rm iso}$ and inserting the leading term of Eq. (C1) gives

$$\begin{aligned} u_{\rm iso}^{\rm max}(0^-) \simeq{}& (98\,{\rm m/s})\frac{n_B^{(+)}}{n_B(0^-)} \\ &\times \left(\frac{\mu_B^{(+)}}{1200\,{\rm MeV}}\right)^{\frac{17}{6}} \left(\frac{10\,{\rm MeV}}{T}\right)^{\frac{5}{6}} \\ &\times \frac{A_{\rm iso}\sqrt{A_{\rm iso}^2 + 162\pi^2T^2/(\mu_B^{(+)})^2}}{\alpha_s^{5/6}\sqrt{1-A_{\rm iso}}\sqrt{1+\xi A_{\rm iso}}}\,. \end{aligned} \tag{34}$$

Writing the average chemical potential as $\mu_B(0^+) + n_K(0^+)\mu_K(0^+)/n_B(0^+)$ and Taylor-expanding the quark matter EoS in Eq. (A4) around its beta-equilibrated state,[MGA Didn't we already do that in Appendix A? Can't we just cite an equation from there?][JS The EoS expansion is now cited above and substituted below.] we can obtain an approximate formula [MGA Are there additional approximations beyond the Taylor expansion?][JS yes: $n_B(x>0) = n_B^{(+)}$ and $\mu_B(x>0) = \mu_B^{(+)}$] that in the parameter range that we explored agrees with the numerical result to within 1%,

$$A_{\rm iso} \approx 3\sqrt{\left(\frac{\mu_B(0^-)}{\mu_B^{(+)}} - 1\right)\frac{(\mu_B^{(+)})^2 + 3\pi^2T^2}{(\mu_B^{(+)})^2 + 9\pi^2T^2}}\,. \tag{35}$$

We can also estimate the thickness $L_{\rm conv}$ of the conversion region, which determines the range of curvature that is compatible with our assumption of a planar geometry. If the boundary moves slowly then when $a \ll 9\pi T/\mu_B^{(+)}$, which is guaranteed to occur at large x where $a \to 0$, the K current equation reduces to

$$\frac{d^2a}{dx^2} = \frac{\gamma_K}{D_K}\left(\frac{9\pi T}{\mu_B^{(+)}}\right)^2 a\,. \tag{36}$$

At large x the temperature reaches an asymptotic value $T(\infty)$ (equal to T in the isothermal case), so we can estimate the thickness of the conversion region as

$$\begin{aligned} L_{\rm conv} &\approx \frac{\mu_B^{(+)}}{9\pi T(\infty)}\sqrt{\frac{D_K}{\gamma_K}} \\ &= (0.2\,\mu{\rm m})\,\frac{1}{\alpha_s^{5/6}}\left(\frac{1200\,{\rm MeV}}{\mu_B^{(+)}}\right)^{7/6}\left(\frac{10\,{\rm MeV}}{T(\infty)}\right)^{11/6}, \end{aligned} \tag{37}$$

where we evaluated D_K using (C1) with $T = T(\infty)$ and γ_K from Eq. (B3). This supports our use of the planar approximation, since the typical curvature of a quark/nuclear boundary in neutron stars will be of order kilometers, much larger than the thickness of the boundary.

IV. BOUNDARY PROPAGATION SPEED: ADIABATIC CASE

A. Adiabatic hydrodynamic equations

The adiabatic limit is obtained by setting thermal conductivity $\kappa_{\rm th} = 0$ and radiative energy loss $Q_\nu = 0$ in Eq. (16). Dropping terms of order u^2 we obtain the adiabatic steady-state equations [JS the numerical code still includes full relativistic corrections,]

$$\frac{d}{dx}(un_B) = 0\,, \tag{38a}$$

$$\frac{d}{dx}(hu) = 0\,, \tag{38b}$$

$$\frac{d}{dx}(\underline{hu^2}+P) = 0\,, \tag{38c}$$

$$\frac{d}{dx}J_K = -\Gamma_K\,. \tag{38d}$$

where $h = P+\epsilon$ is the enthalpy. The first three equations impose constant baryon flux, enthalpy flux and pressure. The upstream values $n_B(0^-)$, $T(0^-)$, and $u(0^-)$ fix these fluxes for the whole solution. Eq. (38d) leads to the same K transport equations as in the isothermal case (18), with the same boundary conditions (Sec. III B). The difference is that in the evolution procedure described in Sec. III A we now determine $T(x)$ from the constant enthalpy flux.

B. Analytical approximation

[MGA We should say explicitly what we keep constant. I made an attempt...] To obtain an approximate analytic expression for the propagation speed of the boundary we follow a similar approach to that of Sec. III C. We make the approximation that as well as the conserved fluxes mentioned above, baryon density $n_B^{(+)}$, baryon chemical potential $\mu_B^{(+)}$, and fluid speed $u^{(+)}$ are independent of x in the quark matter region. Temperature and K fraction a are allowed to vary with x. We can then follow the same reasoning as in Sec. III C to obtain an analytic estimate of the speed in terms of the integrals I_1 (28) and I_2 (29).

As in the isothermal case, we parameterize I_1 in terms of a sharpness factor ξ to obtain the same expression for the speed in terms of I_2, Eq. (31). To evaluate I_2, we need to know how the K diffusion coefficient D_K and K creation rate Γ_K vary through the conversion region. In our approximation the pressure and baryon chemical potential have constant values $P^{(+)}$ and $\mu_B^{(+)}$ in that region, so the system follows an isobar in the μ_K-T plane. We will expand the quark matter EoS to second order in T/μ_B and μ_K/μ_B, Eq. (A5), which is a good approximation for $a \ll 1$ but becomes less accurate as a approaches 1, where μ_K/μ_B can be of order 0.5. Then the isobar takes the form of a quadrant of an ellipse in the μ_K-T plane. Since to this order there is a linear relationship between the K fraction a and μ_K (Eq. (A8)) the quark matter also follows an elliptical quadrant in the a-T plane,

$$T^2 = \frac{2(\mu_B^{(+)})^2}{81\pi^2}(A_{\rm adb}^2 - a^2) \tag{39}$$

as shown by the dashed line in Fig. 4. Here $A_{\rm adb}$, the value of a where the quadratic approximation to the isobar (dashed line) meets the $T = 0$ axis in Fig. 4, is defined by $P_{\rm QM}(A_{\rm adb}, T = 0) = P^{(+)}$, so to leading order, [MGA in what?][JS Quark EoS, in $\frac{\mu_K}{\mu_B}$ and $\frac{T}{\mu_B}$]

$$A_{\rm adb} = (\approx?)\frac{3}{2}\frac{\sqrt{108\pi^2\left(P^{(+)} + U_{\rm bag}\right) - \left(\mu_B^{(+)}\right)^4}}{\left(\mu_B^{(+)}\right)^2}. \tag{40}$$

Using Fig. 4 we can describe the conversion region as follows. At $x = 0^+$ the quark matter has some initial K fraction $a(0^+)$ which determines the initial temperature $T(0^+)$ (the red star in the figure). As x increases, the system moves upwards along the elliptical isobar, ending at $a = 0$ and some final temperature $T(\infty)$ when $x \to \infty$ (the red square in the figure). We see from the figure that the analytical approximation (dashed line) gives a reasonably accurate approximation to the actual path followed in the (a, T) plane.

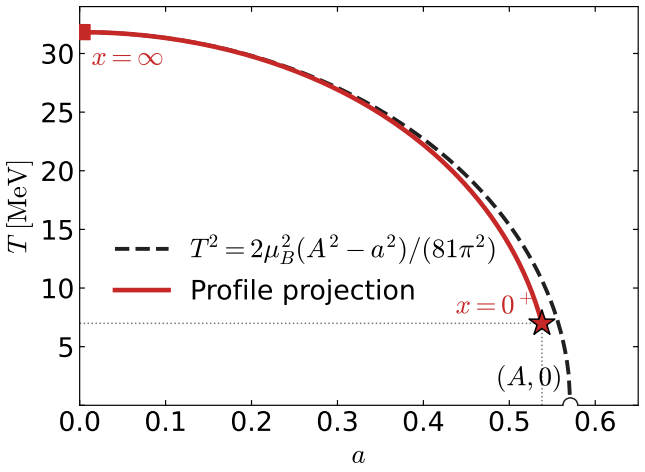


FIG. 4: Constant-pressure trajectory through the quark conversion layer in the adiabatic case, for $T(0^-) = 10$ MeV, $B^{1/4} = 180$ MeV, $n_B(0^-) = 3n_0$, and $a(0^+) = 0.55$. The red curve is from the profile obtained by numerically solving the K flux equation. The dashed curve is the quarter ellipse from the analytic approximation Eq. (39). [MGA The detailed description can be in the main text. The dashed curve should be labeled "analytic approx" and the red curve should be labeled "numerical solution". The place where the dashed curve meets the x axis should be labeled $A_{\rm adb}$.]

We can now evaluate the integral I_2 (29) by using the results from Appendices B and C) for $\Gamma_K(a, T)$ and $D_K(a, T)$ (keeping only the first term), and using (39) to express T as a function of a. Setting $a(0^+) = A_{\rm adb}$ we find

$$I_2 = \frac{27\pi^{7/3}}{56\sqrt{2}}\frac{A_{\rm adb}^{7/3}\gamma_K}{\alpha_s^{5/3}h_D\mu_B^{(+)}} \times \left[8 + \left(1 - \frac{a(0^+)^2}{A_{\rm adb}^2}\right)^{1/6}\left(\frac{a(0^+)^2}{A_{\rm adb}^2} - 8\right)\right]. \tag{41}$$

Unlike the isothermal result in Eq. (32), here $D_K[T(a)]$ remains inside the integral and produces the $A_{\rm adb}^{7/3}$ prefactor and the fractional powers.

So from (31),

$$u_{\rm adb}^2(0^-) = \frac{9\pi^{7/3}}{4\sqrt{2}\,h_D\alpha_s^{5/3}}\frac{\gamma_K A_{\rm adb}^{7/3}}{\mu_B^{(+)}\lambda_n^2\left(1 - a(0^+)\right)\left(1 + \xi a(0^+)\right)} \times \left[\frac{24}{7} - 3\left(1 - \frac{a^2(0^+)}{A_{\rm adb}^2}\right)^{1/6} - \frac{3}{7}\left(1 - \frac{a^2(0^+)}{A_{\rm adb}^2}\right)^{7/6}\right]. \tag{42}$$

Our framework does not predict the value of $a(0^+)$, so as in the isothermal case we will choose the value that gives the highest boundary propagation speed. Eq. (42) shows that the speed of the boundary grows as $a(0^+)$ gets larger. The factor in square brackets has a weak dependence on $a(0^+)$, while the denominator $\big(1-a(0^+)\big)\big(1+\xi a(0^+)\big)$ is a decreasing function of $a(0^+)$ for every $-1<\xi<1$, as in Sec. III C. To obtain the maximum boundary speed $u_{\rm adb}^{\rm max}(0^-)$ we therefore set $a(0^+)$ to the largest physically possible value, which we see from Fig. 4 is $A_{\rm adb}$, since this corresponds to the lowest allowed initial quark matter temperature, $T(0^+)=0$.

Physically it makes sense that larger $a(0^+)$gives faster propagation of the boundary because it corresponds to a lower strangeness fraction in the freshly created quark matter, closer to the flavor content of the neutron matter, so it is in some sense easier to create.

Setting $a(0^+)=A_{\rm adb}$ in (42) we find the maximum speed

$$u_{\rm adb}^{\rm max}(0^-)=(82\,{\rm ms}^{-1})\,\frac{n_B^{(+)}}{n_B(0^-)}\left(\frac{\mu_B^{(+)}}{1200\,{\rm MeV}}\right)^2\times \frac{A_{\rm adb}^{7/6}}{\alpha_s^{5/6}\sqrt{1-A_{\rm adb}}\sqrt{1+\xi A_{\rm adb}}}\,. \tag{43}$$

Thus the typical speed of the boundary is expected to be of order $100{\rm m\,s}^{-1}$. In this expression the only factor that could drive the speed to a significantly higher value is $1/\sqrt{1-A_{\rm adb}}$. To see when $A_{\rm adb}$ could approach 1 we recall that $A_{\rm adb}$ is the largest possible K fraction in the freshly created quark matter at $x=0^+$. The K fraction in neutron matter is 1, so if $A_{\rm adb}=1$ then the pressure at the boundary is so high that that nuclear matter can convert to quark matter with the same flavor content. In this case no weak interactions are needed, so it is not surprising that at such high pressures our approach breaks down.

[MGA Put this somewhere] When we solve the profile equations numerically we will find that in the adiabatic case $\xi=-0.85$ is a good approximation (Appendix D). [MGA Is this still the best value?][JS-0.85 is better than -0.5. I'll try some other values and see which one is the best. – I've tried different ξ and for the two analytic method to agree well with full numerical on average, $\xi=-0.865$ is better, but only a marginal improvement from -0.85.]

V. RESULTS

Do we need to re-calculate and re-produce every plot?

- use quark EoS without gluons [MGA No, as long as we can cite reputable sources for the EoS with gluons]
- use full phase-space integral rate Γ_K for full numerical calculations [MGA I don't think we decided on this]
- first, the set up

Need to rewrite this intro so it is not tied to the adiabatic computation, and just gives the essential information for understanding our results. Maybe some of the material can be moved to where we discuss the adiabatic results.

We discuss the energy conserving front system Eq. (38) first. We choose the bag constant by requiring cold neutron matter and cold equilibrated strange quark matter to coexist at neutron matter critical baryon number density $n_B^{\rm crit}(T{=}0)=3n_0$. This gives $U_{\rm bag}^{1/4}=189.2\,{\rm MeV}$. Since we are not including Urca processes, we set strange quark mass to zero. We also adopt the conventional phenomenological choice $\alpha_s=0.3$ for bag-model quark matter [35, 47]. For each pair of upstream variables $n_B(0^-)$ and $T(0^-)$, we evolve the K density and current, reconstruct the local fraction $a(x)=n_K(x)/n_B(x)$, and enforce the conserved baryon, momentum, and energy fluxes across the quark phase.

But to get a unique solution to Eq. (38), hence a unique value for $u(0^-)$, we still need a proscribed $a(0^+)$, or equivalently, $T(0^+)$. As mentioned in Sec. IV B, we can utilize this freedom to maximize the velocity by choosing $T(0^+)=0$, but practically that would make $D_K^{-1}=0$ hence giving us a non-Lipschitz initial condition for the ODE system Eq. (18), which gives us a family of possible solutions instead of one unique solution. So in practice, when using analytical approximation Eq. (31) or Eq. (42) we can set $a(0^+)=A_{\rm adb}$ or $T(0^+)=0$, but when numerically solving the full system Eq. (38) we need to set a small but non-zero $T(0^+)$ to avoid the non-Lipschitz initial condition. From our numerical results Fig. 7, we find that the velocity doesn't change over two times for $T(0^+)<1$ MeV, so we choose $T(0^+)=0.1$ MeV as a representative value for the numerical results.

A. Phase boundary speed

- about contour plot:
  - existence of stable neutron matter region, and why it takes that shape
  - $A_{\rm adb}>1$ region and why that shape
  - velocity in between, also mention it diverge as the input goes close to $A_{\rm adb}=1$

AH I think if we have Fig1 as schematic, we should have an equivalent figure here with an actual result?

VI. CONCLUSION

- Impact of instabilities, Urca processes, ... on phase boundary speed

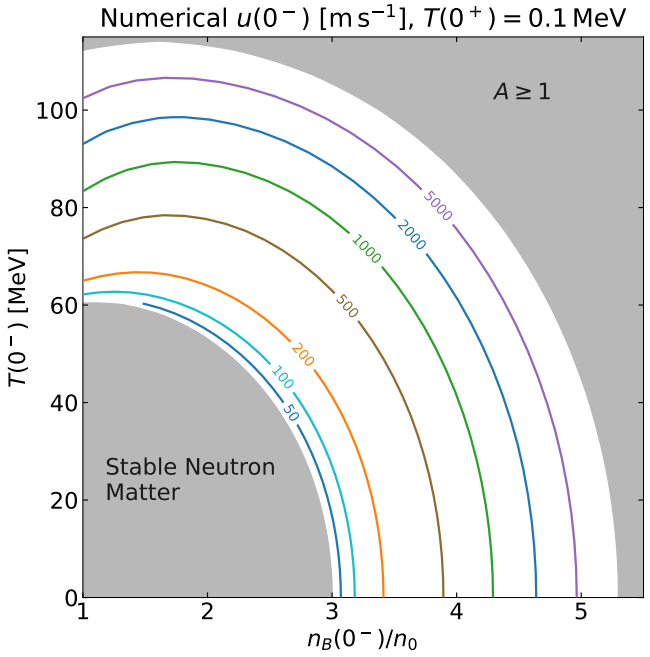


FIG. 5: Full numerical phase-boundary speed $u(0^-)$ in $\mathrm{m\,s^{-1}}$ as a function of the upstream baryon density and temperature, with $T(0^+) = 0.1\,\mathrm{MeV}$ and $n_B^{\mathrm{crit}}(T{=}0) = 3n_0$. The white band is the domain in which the moving-front solution exists. Below the lower boundary neutron matter is stable against the assumed conversion, while the region with $A_{\mathrm{adb}} \geq 1$ lies outside the weak-interaction-limited branch described by our model. The contours are drawn at 50, 100, 200, 500, 1000, 2000, and $5000\,\mathrm{m\,s^{-1}}$.

Our results allow us to speculate on the fate of interface modes in the inspiral. Typical frequencies of interface modes depend on the EoS of nuclear and quark matter as well as the star's mass, where heavier stars show lower frequency resonances. The frequencies themselves range from a couple hundred to around $f = 2000$ Hz (see Fig. 2 of Ref. [1]). Depending on the energy absorbed by the mode, the amplitude ranges from a few millimeters to centimeters in the inspiral, and might reach several meters in the postmerger. Assuming a frequency of $f = 700$ Hz, we can calculate the resonant velocity where the timescale of the oscillation matches the time given by the amplitude divided by the phase boundary velocity. In this resonant scenario, we expect viscous effects as described in [25] become important. At even larger velocities, we expect the mode to lose its support (see discussion in the nuclear case in Ref. [48]) and vanish. In the inspiral, the resonant velocity ranges from $v_{\mathrm{res}} = 1$ m/s to $v_{\mathrm{res}} = 15$ m/s, which is within the range found in our calculation shown in Fig. 5. In the postmerger scenario, the resonant velocity rises to $\mathcal{O}(1000\,\mathrm{m/s})$, which is reached in the strongly metastable scenario (see Fig. 5). This shows that any accurate calculation of the interface mode must include the finite phase boundary velocity and viscous effects, potentially via the dynamical sound-speed formalism developed in Ref. [48] and applied to quark matter in [cite Zhao Han Haber].

This also motivates a more detailed calculation of the phase boundary velocity:

- In this paper, we model the baryonic side as pure neutron matter. However, in compact stars, baryonic matter is composed of chemically equilibrated nuclear matter, which includes not only neutrons, but also protons, electrons, and muons in a composition determined by charge neutrality and chemical equilibration. Even hyperonic matter could be considered on the baryonic side of the phase boundary.

- The presence of leptonic particles opens up the possibility of proton or strangeness-changing Urca rates. The phase boundary velocity would then potentially be slowed down even further, as we now need to equilibrate multiple particle fractions on the quark matter side via the generally slower Urca processes. Our results thus likely present an upper bound on the velocity.

- At high temperatures, the physics of neutrinos might become important as neutrinos could cool down the conversion region, or if they get trapped, will alter the hydrodynamic description of the phase boundary. However, given our recent work in Refs. [49, 50], neutrino trapping on the microscopic scale of the phase boundary seems unfeasible, but could potentially play a role in the equilibrated baryonic or quark matter

- At the temperatures occurring in the inspiral, baryonic and quark matter might be found in a (color-)superconducting/superfluid state. Cooper pairing will not only alter the hydrodynamics of the problem, but suppresses the flavor-changing rates used in this work, and will also alter diffusive transport, thus furder slowing down the phase boundary.

- In this work, we have assumed that thermal equilibration across the phase boundary is happening on a timescale slower than the evolution of the phase boundary. Towards the end of the equilibration process, however, the phase boundary slows down ([AH]do we have a section explaining this?. To not only reach flavor, but also thermal equilibration, thermal equilibration across the boundary has to be taken into account. Otherwise, the final state of the evolution will be wrongly described as a state where the baryonic and the quark matter phase stay at different temperatures.

- In this work, we assume that all profiles on the nuclear matter side are constant, and neglect any disturbance propagating ahead of the phase boundary.

- Furthermore, the assumption of a steady state, which we deem reasonable, should be further investigated.

- Our hydrodynamic description is based on an ideal fluid description. Dissipative corrections might be

taken into account in the stress-energy tensor, or be included via weak interactions as described in Ref. [51].

- If the phase boundary is driven by large-scale oscillations (as we expect it in the inspiral and the merger), the opposite scenario, where quark matter recombines to nuclear matter, has to be calculated. This will also allow us to estimate the strength of the so-called "phase conversion dissipation", a dissipative effect stemming from the out-of-phase movement of the actual phase boundary with the critical pressure [25].

- Cooper pairing: influence on diffusion, weak interaction rates, anything else?

- surface tension?

ACKNOWLEDGEMENTS

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Appendix A: Bag model of quark matter

In the approximation that we use for quark matter, neglecting electromagnetic interactions and leptons, we impose quark charge neutrality, so $n_u = n_B$, $n_d = n_B(1+a)$, and $n_s = n_B(1-a)$. The pressure is the sum of the quark and gluon contributions with a deconfined vacuum energy offset (bag constant) [47, 52],

$$\begin{aligned} P_{\rm QM}(n_B, a, T) &= P_q(n_B, T) + P_q\big(n_B(1+a), T\big) \\ &\quad + P_q\big(n_B(1-a), T\big) + \frac{8\pi^2}{45}T^4 - U_{\rm bag}\,, \end{aligned} \tag{A1}$$

where $P_q(n_f, T)$ is the pressure of a free massless Fermi gas of three colors of a single quark flavor at net number density n_f,

$$P_q(n_f, T) = \frac{\mu_f^4}{4\pi^2} + \frac{\mu_f^2 T^2}{2} + \frac{7\pi^2 T^4}{60}\,, \tag{A2}$$

$$n_f = \frac{\mu_f^3}{\pi^2} + \mu_f T^2\,. \tag{A3}$$

These expressions are exact for free massless quarks and antiquarks, with two spin states and three colors, at arbitrary μ_f/T [53, Eq. (7.42)]. The density relation fixes each μ_f, with $\mu_B = \mu_u + \mu_d + \mu_s$ and $\mu_K = \mu_d - \mu_s$. We expect $\mu_K, T \ll \mu_B/3$ so we expand Eq. (A1) in powers of these at fixed μ_B to obtain

$$\begin{aligned} P_{\rm QM} = -U_{\rm bag} &+ \frac{\mu_B^4}{108\pi^2} + \frac{\mu_B^2}{12\pi^2}\mu_K^2 - \frac{\mu_K^4}{32\pi^2} \\ &+ \frac{\mu_B^2}{6}T^2 + \frac{1}{4}\mu_K^2 T^2 + \frac{19\pi^2}{36}T^4 + \cdots\,. \end{aligned} \tag{A4}$$

In our analytic approximations we will expand the equation of state to second order in μ_K and T,

$$P_{\rm QM} = -U_{\rm bag} + \frac{\mu_B^4}{108\pi^2} + \frac{\mu_B^2}{12\pi^2}\mu_K^2 + \frac{\mu_B^2}{6}T^2\,. \tag{A5}$$

Differentiating Eq. (A5) gives

$$n_K = \left.\frac{\partial P_{\rm QM}}{\partial \mu_K}\right|_{\mu_B, T} \simeq \frac{\mu_B^2 \mu_K}{6\pi^2}\,, \tag{A6}$$

$$n_B = \left.\frac{\partial P_{\rm QM}}{\partial \mu_B}\right|_{\mu_K, T} \simeq \frac{\mu_B^3}{27\pi^2}\,. \tag{A7}$$

We can also express the quark matter properties in terms of the K fraction a. To leading order,

$$a \equiv \frac{n_K}{n_B} \simeq \frac{9\mu_K}{2\mu_B}\,. \tag{A8}$$

Appendix B: Strangeness creation

For positive μ_K, the nonleptonic process relaxes the K density at the rate [54]

$$\Gamma_K(\mu_B, \mu_K, T) = \mathcal{C}_K \mu_u^5 \mu_K \left(\mu_K^2 + 4\pi^2 T^2\right)\,, \tag{B1}$$

[MGA Are we using Madsen (21) or Madsen (22)? (21) is for the limit $\mu_K = 0$ and $\mu_u = \mu_d$ which is not what we assume in our bag model.][JS We using Madsen Eq. (23)] where

$$\mathcal{C}_K = \frac{16}{5\pi^5} G_F^2 \sin^2\theta_C \cos^2\theta_C\,, \tag{B2}$$

where $G_F = 1.16637 \times 10^{-11}\,\mathrm{MeV}^{-2}$ is the Fermi constant and $\theta_C = 13.02°$ is the Cabibbo angle. We use $\mu_u = \mu_B/3 + 2\mu_Q/3$, where electric charge chemical potential μ_Q is fixed by charge neutrality, $2n_u - n_d - n_s = 0$.

For the analytic approximations in Secs. III C and IV B, we use the quadratic approximation of Sec. A to express the rate in terms of the K fraction. Using (A8) and $\mu_u \simeq \mu_B/3$, we obtain

$$\begin{aligned} \Gamma_K(a, T) &= n_B \gamma_K \left(a^3 + \frac{81\pi^2 T^2}{\mu_B^2} a\right)\,, \\ \gamma_K &= \frac{128}{3^8 \times 5\pi^3} G_F^2 \sin^2\theta_C \cos^2\theta_C\, \mu_B^5\,. \end{aligned} \tag{B3}$$

Appendix C: Strangeness diffusion

The diffusion of the K flavor in quark matter is described by a diffusion coefficient D_K, which, since the quarks are relativistic, is closely related to the mean free path. This is determined by strong interactions, so we model it using the single gluon exchange perturbative expression [46, 55]

$$D_K^{-1} = \frac{8N_f\alpha_s^2}{\pi}\left(h_D\frac{T^{5/3}}{q_D^{2/3}} + \frac{\pi^3T^2}{12q_D}\right), \qquad \text{(C1)}$$

Here $N_f = 3$ is the number of active quark flavors, $\alpha_s = 0.3$ is the strong coupling, $h_D = \Gamma(\frac{8}{3})\zeta(\frac{5}{3})(2\pi)^{2/3} \simeq 1.81$ is a numerical prefactor, q_D is the Debye wavenumber defined by $q_D^2 = g^2N_f\mu_B^2/(18\pi^2)$, and $g^2 = 4\pi\alpha_s$. In our numerical calculations we set $\alpha_s = 0.3$ [MGA Can we find some standard citation about the running of the QCD coupling that would show that this is an appropriate value at energy scales of order 400 MeV?][35, 47] .

In (C1) the first term comes from the interaction mediated by Landau-damped transverse gluons, and the second term comes from Debye-screened longitudinal [MGA and transverse?] gluons. The first term is larger at temperatures up to $\sqrt{\alpha_s}\mu_B/6$, which is of order 70 MeV, but because both terms have similar T dependence they are already comparable (within a factor of 2) when the temperature rises above $0.04\sqrt{\alpha_s}\mu_B/3$, which is of order 7 MeV. In neutron star mergers the temperature can range from below 1 MeV (early inspiral) to about 70 MeV, so for our numerical calculations we use the full expression, but for the analytic approximations of Secs. III C and IV B we only use the Landau damping term.

AH shall I add the derivation of K diffusion from individual quark currents that I have typed up in the notes?

Appendix D: Shape of the diffusive flux profile

The diffusive flux $g(x) \equiv -D_K(x)a'(x)$ enters I_1 (Eq. (28)) only through its shape as a function of a rather than x: since $a(x)$ is monotonic, we can trade x for a and write $I_1 = \int_0^{a(0^+)} g(a)\,da$, the area under g between the equilibrated downstream value $a = 0$, where $g = 0$, and the interface value $a = a(0^+)$, where the jump condition Eq. (26) fixes $g(a(0^+)) = \lambda_n u(0^-)[1 - a(0^+)]$.

Write $G \equiv g(a(0^+)) = \lambda_n u(0^-)[1 - a(0^+)]$ for this interface value. Three limiting shapes bracket the possibilities for the area under g. If g rises linearly from 0 to G over $0 \le a \le a(0^+)$, the area is a triangle, $I_1 = \frac{1}{2}Ga(0^+)$. If g instead jumps to G immediately and stays there, the area is the full rectangle, $I_1 = Ga(0^+)$. If g instead stays near 0 over almost the whole range and only rises to G at the very end, the area shrinks toward 0. These are the $\xi = 0$, $\xi = 1$, and $\xi = -1$ limits of

$$I_1 = \frac{1+\xi}{2}\,Ga(0^+)\,, \qquad \text{(D1)}$$

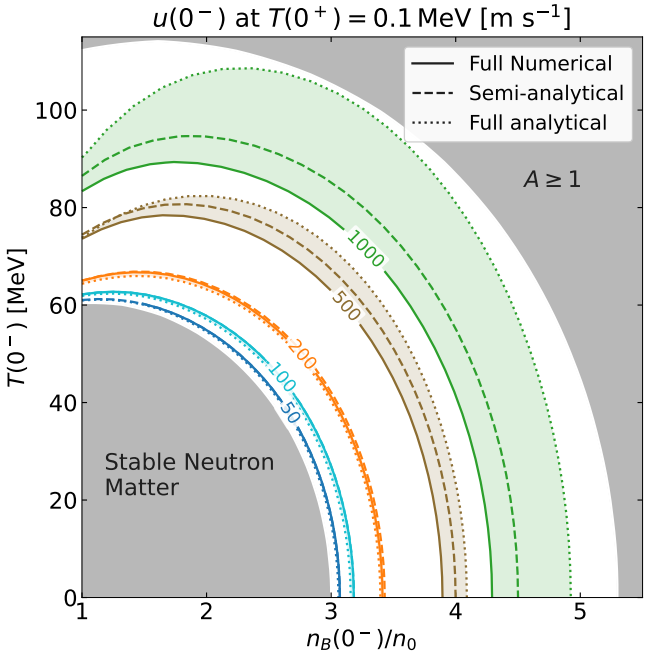


FIG. 6: Comparing three methods for $T(0^+) = 0.1$ MeV.

which is Eq. (30) written in terms of G: ξ measures whether the diffusive flux builds up early ($\xi > 0$) or late ($\xi < 0$) across the conversion region, relative to a linear ramp ($\xi = 0$). For each input $T(0^-)$ and $n_B(0^-)$, there is one unique ξ that makes Eq. (30) exact.

The full numerical solutions to Eq. (18) are smooth between the two endpoints, and after comparing the plots of $g(a)$ using profiles that numerically solve Eq. (38), we find that $\xi = -0.5$ is a good approximation, meaning that $g(a)$ is mostly convex. We use $\xi = -0.5$ in Eq. (31) and Eq. (42) throughout.

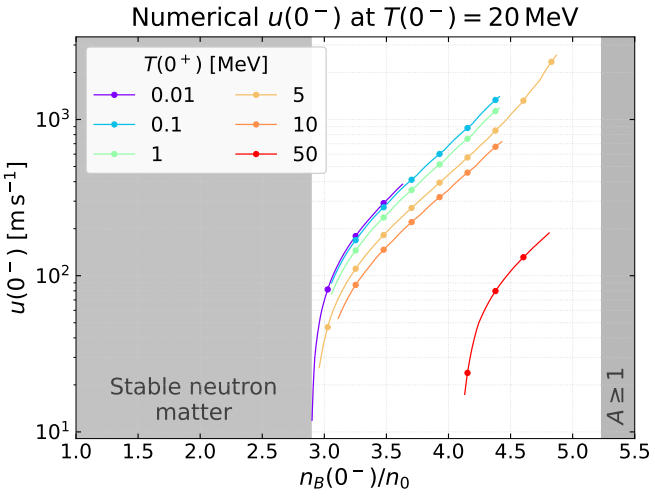
Appendix E: Numerical method

[MGA If we want to keep this appendix then we should flesh it out with some info about how to solve the equations, e.g. about shooting with trial values of $u(0^-)$ to find the right value.]

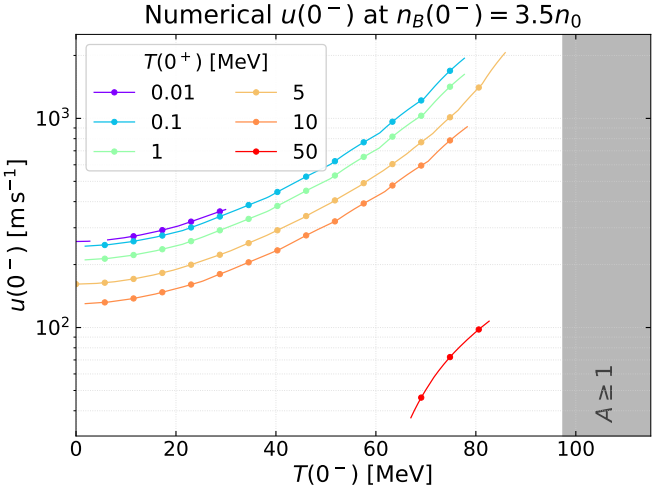
We solve Eq. (18) at fixed temperature in the isothermal case, or at fixed energy flux [Eq. (38b)] in the adiabatic case. We compactify the semi-infinite quark region using

$$x = -L_0\ln(1-s)\,, \qquad 0 \le s \le 1 - \epsilon_{\rm tail}\,, \qquad \text{(E1)}$$

where L_0 is fixed from the initial interface diffusion–reaction scale. The result is checked for stability under changes of both L_0 and $\epsilon_{\rm tail}$. We first attempt the requested $a(0^+)$ directly; if necessary, we continue from a smaller interface fraction and use each converged profile as the initial guess for the next step. States with nega-



(a)



(b)

FIG. 7: Dependence of the numerical phase-boundary speed on the prescribed interface temperature $T(0^+)$. Panel (a) shows $u(0^-)$ as a function of $n_B(0^-)$ at fixed $T(0^-) = 20\,\mathrm{MeV}$, while panel (b) shows $u(0^-)$ as a function of $T(0^-)$ at fixed $n_B(0^-) = 3.5n_0$. The velocity axis is logarithmic, the markers denote the numerical slice samples, and the line color runs from colder (purple and blue) to hotter (red) values of $T(0^+)$. The gray regions are excluded in the corresponding slice by stable neutron matter or by $A_{\mathrm{adb}} \geq 1$.

tive temperature, nonpositive density, a failed equation-of-state closure, or nonphysical transport coefficients are rejected. For the isothermal and adiabatic cases, these profiles provide the numerical reference against which the analytic results in Eqs. (33) and (42), respectively, are tested.

Appendix F: Local thermal equilibrium

[LB We might be able to turn this into a sentence in the main body.]However, $T(0^+) = 0$ lies outside the validity of the present diffusion treatment. As $T(0^+) \to 0$, the quark mean free path λ_Q^{mfp} diverges, so quark matter cannot remain in local thermal equilibrium (LTE) across a conversion layer with a finite variation scale. Requiring the mean free path to remain shorter than the scale on which the diffusion coefficient changes,

$$\lambda_Q^{\mathrm{mfp}} = 3D_K < \frac{D_K}{|dD_K/dx|}\,, \tag{F1}$$

and using the constant-pressure trajectory and interface gradient gives

$$5u(0^-)\lambda_n a(0^+)[1-a(0^+)] < A_{\mathrm{adb}}^2 - a^2(0^+)\,. \tag{F2}$$

The LTE-limited composition is obtained by saturating this inequality,

$$A_{\mathrm{adb}}^2 - \left[a^{\mathrm{LTE}}(0^+)\right]^2 = 5\lambda_n u(0^-) a^{\mathrm{LTE}}(0^+)\left[1 - a^{\mathrm{LTE}}(0^+)\right], \tag{F3}$$

with $u(0^-)$ evaluated at $a(0^+)$ from Eq. (42).

Finally, the full numerical solutions confirm the scale estimate: across the conversion layer, the relative variations of n_B and P are typically no larger than 10^{-10} and 10^{-12}, respectively. The analytic approximation removes variation that is numerically invisible while retaining the composition and temperature profiles that set the conversion rate.

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